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5 **(a)** Atmospheric pressure is 1.0×10^5 Pa. A volume of $1.02\,m^3$ of air at atmospheric pressure is pumped into a car tyre. The pressure of the air when it is inside the tyre is 2.6×10^5 Pa.

Calculate the volume of the air when it is in the car tyre.

volume = ... m^3 [3]

(b) **(i)** Describe the structure of a gas in terms of the separation, arrangement and motion of particles.

separation .....

.....

arrangement .....

.....

motion .....

.....

[3]

(ii) Explain, in terms of particles, why the internal energy of the gas decreases when the gas is at a lower temperature.

.....

..... [2]

[Total: 8]

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